



Spin Waves in Systems with Long Period Helical Spin Density Waves Due to the Antisymmetric and Symmetric Exchange Interactions

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In order to elucidate the different behaviours of the helical spin density waves (HSDW) due to the antisymmetric and symmetric exchange interactions, spin waves in the conical spin density wave (CSDW) states and the induced ferromagnetic states (IFMS) under external magnetic fields are calculated by use of phenomenological Hamiltonians with and without the antisymmetric exchange interaction. It is found that the spin wave spectra for the two types of HSDW are strongly different from each other, especially in the magnetic field dependences, enough to be able to distinguish them. The ESR frequencies are also investigated theoretically for both types. In the light of these results, the observed magnetic field dependences of the ESR frequencies in MnSi and $\text{Fe}_{1-x}\text{Co}_x\text{Si}$ are discussed to prove the HSDW due to the antisymmetric exchange interaction in these compounds.

§1. Introduction

Until now, the two types of helical spin density wave (HSDW) which are different in their origins have been known. The one is the HSDW due to the symmetric exchange interaction (SEI), i.e., the HSDW arising as the result of a competition between some long-ranged SEI's. It is well established that most of the HSDW's observed in transition-metal compounds and rare-earth compounds belong to this type. The other type of HSDW is realized by the presence of the antisymmetric exchange interaction (AEI). This types of HSDW is confirmed first theoretically¹⁾ and experimental²⁻⁴⁾ in MnSi with the B20 crystal structure which has no inversion center. At present, the HSDW's in $\text{Fe}_{1-x}\text{Co}_x\text{Si}$,⁵⁻⁷⁾ FeGe ,⁸⁾ $\text{Cr}_{1/3}\text{NbS}_2$,^{9,10)} and CsCuCl_3 ¹¹⁾ are also considered to belong to the latter type. The HSDW due to the AEI has some characteristic properties such as long periods and the occurrence of only one of the clockwise and counter-clockwise rotations of spins. However, not only these properties but also their thermodynamical behaviours can be scarcely direct evidences of the HSDW due to the AEI, since the HSDW due to the SEI also can often exhibit those properties in extraordinary

cases.¹²⁾ The purpose of this paper is to show that the spin waves in the two types of HSDW are essentially different from each other and therefore they can always identify the origins of HSDW's observed in real compounds.¹³⁾

The spin wave in the HSDW due to the AEI was discussed in 1970 on the basis of a supposed Hamiltonian by Bar'yakhter and Stefanovsky.¹⁴⁾ After the discovery of the HSDW due to the AEI in MnSi in 1980, the author calculated independently the spin wave spectra by use of the Hamiltonian for MnSi to find the characteristic behaviours of the spin wave under external magnetic fields.¹³⁾ An extension of the calculation, which takes account of magneto-elastic effects, was made by Plumer.¹⁵⁾ In this paper, we investigate in detail the spin wave energies of both systems with the different types of long-period HSDW's in the conical spin density wave (CSDW) states and the induced ferromagnetic states (IFMS) under external magnetic fields.

Since the period of the HSDW due to the AEI is very long, e.g., 180 Å in MnSi²⁾ and 350 Å in $\text{Fe}_{0.8}\text{Co}_{0.2}\text{Si}$,⁵⁾ the magnitude of the propagation vector $\mathbf{Q}$ is much small compared with that of a reciprocal lattice vector. For such a small $|\mathbf{Q}|$, neutron inelastic scattering intensities can not be necessarily measured

with high accuracies enough to reveal behaviours of spin waves with wave vectors near $|\mathbf{Q}|$. On the other hand, Cooper *et al.*¹⁶⁾ had shown for the HSDW due to the SEI that the energies of the spin waves with the wave vectors $\mathbf{q}=0$, and $\pm\mathbf{Q}$ can be observed by making electron spin resonance (ESR) experiments. Date *et al.*¹⁷⁾ and Watanabe *et al.*⁶⁾ could measure the ESR frequencies in MnSi and $\text{Fe}_{1-x}\text{Co}_x\text{Si}$, respectively. In order to obtain an evidence of the origin of the HSDW's, the ESR frequencies under static magnetic fields are investigated to be compared with the experimental results.

In the next section, a phenomenological Hamiltonian is introduced to give a HSDW due to the AEI and the magnetic field effect on the HSDW. After the equilibrium spin structure is discussed briefly, the spin wave energy is calculated by means of the macroscopic spin wave theory. The parallel arguments are made for the case of the HSDW due to the SEI in

§3. Section 4 is devoted to the investigation of the ESR frequencies in both types of HSDW. Finally, concluding remarks are given in §5.

§2. Spin Waves in the System with the Helical Spin Density Wave (HSDW) Due to the Antisymmetric Exchange Interaction (AEI)

2.1 Phenomenological Hamiltonian and ground spin structure

Spin systems whose spin densities $\mathbf{S}(\mathbf{r})$ vary slowly with the position $\mathbf{r}$ can be unstable against the HSDW in 18 types of crystal structure with no inversion center because of the presence of the AEI.¹²⁾ Among them, we consider here spin systems in crystals with the isogonal point groups T and O , because such a spin system is found in B20 ($P2_13(T^4)$)-type compounds including MnSi and $\text{Fe}_{1-x}\text{Co}_x\text{Si}$. By taking account of the crystal symmetry, our Hamiltonian for a unit volume is chosen to be^{12,13)}

$$\mathcal{H} = \int d\mathbf{r} \left\{ \frac{c}{2} \sum_{ii'} \left(\frac{\partial S_i(\mathbf{r})}{\partial r_{i'}} \right)^2 - D \mathbf{S}(\mathbf{r}) \cdot [\nabla \times \mathbf{S}(\mathbf{r})] - \frac{1}{2} K_1 \sum_i S_i(\mathbf{r})^4 - \mathbf{H} \cdot \mathbf{S}(\mathbf{r}) \right\}, \quad (1)$$

where $\mathbf{S}(\mathbf{r})$ is the spin density divided by $\hbar$, i and i' denote the coordinate axes x , y , and z referred to the cubic axes, c and D are, respectively, the SEI and AEI constants, K_1 is the cubic magnetic anisotropy constant, and $\mathbf{H}$ is an external magnetic field multiplied by $g\mu_B$ with the gyromagnetic ratio g . In eq. (1), the SEI has been approximated to be isotropic, since its anisotropic part is very small and does not play any important role in the following arguments. As in the previous papers,^{1,12,13)} c is assumed to be positive. Then, eq. (1) with $\mathbf{H}=0$ gives an equilibrium HSDW with the propagation vector $\mathbf{Q}$ which is parallel to $[001]$ for $4D^2/c > K_1 S^2 > 0$ and to $[111]$ for $6D^2/c > -K_1 S^2 > 0$. The positive or negative $Q = -D/c$ corresponds, respectively, to the

counterclockwise and clockwise rotations of $\mathbf{S}(\mathbf{r})$. When $\mathbf{H}$ parallel to $\mathbf{Q}$ is increased, the HSDW is distorted to be a CSDW without changing $\mathbf{Q}$ and finally squeezed into the IFMS above the critical magnetic field H_c .¹²⁾

Here, we define two new coordinate systems, which are shown in Fig. 1. The system axes (X, Y, Z) , which is fixed in the crystal, is chosen so that the Z axis is parallel to $\mathbf{Q}$, while the axes (ξ, η, ζ) , which is rotated along the Z axis, is chosen so that the ζ axis is parallel to the static component of a local spin density vector $\mathbf{S}(\mathbf{r})$. The relations between the three coordinate systems are given in Appendix I. By transforming (x, y, z) into (X, Y, Z) and $(S_x(\mathbf{r}), S_y(\mathbf{r}), S_z(\mathbf{r}))$ into $(S_\xi(\mathbf{r}), S_\eta(\mathbf{r}), S_\zeta(\mathbf{r}))$, eq. (1) is changed within a linear theory for the spin motion to be

$$\begin{aligned} \mathcal{H} = \int d\mathbf{r} \left(\frac{c}{2} \sum_j \left[\left(\frac{\partial S_\xi(\mathbf{r})}{\partial r_j} \right)^2 + \left(\frac{\partial S_\eta(\mathbf{r})}{\partial r_j} \right)^2 \right] \right. \\ \left. + D \sin \theta \left\{ \cos(QZ) \left[S_\xi(\mathbf{r}) \left(\frac{\partial S_\eta(\mathbf{r})}{\partial X} \right) - S_\eta(\mathbf{r}) \left(\frac{\partial S_\xi(\mathbf{r})}{\partial X} \right) \right] \right\} \right) \end{aligned}$$

$$+ \sin(QZ) \left[S_{\xi}(r) \left(\frac{\partial S_{\eta}(r)}{\partial Y} \right) - S_{\eta}(r) \left(\frac{\partial S_{\xi}(r)}{\partial Y} \right) \right] \left\} - \frac{D^2}{2c} [S_{\xi}(r)^2 + \cos^2 \theta S_{\eta}(r)^2 + \sin^2 \theta S_{\zeta}(r)^2] \right. \\ \left. - \frac{1}{2} K_1 [\psi_{\xi}(\theta) S_{\xi}(r)^2 + \psi_{\eta}(\theta) S_{\eta}(r)^2 + \psi_{\zeta}(\theta) S_{\zeta}(r)^2] S_{\zeta}(r)^2 - HS_{\zeta}(r) \cos \theta \right\}, \quad (2)$$

where H has been applied along Q , j denotes X , Y , and Z , θ is the angle between Q and the ζ axis, and $\psi_{\xi}(\theta)$, $\psi_{\eta}(\theta)$, and $\psi_{\zeta}(\theta)$ are defined by

$$\begin{aligned} \psi_{\xi}(\theta) &\equiv (1 + \cos^2 \theta)(1 - \psi) + \frac{3}{2} (1 - \cos^2 \theta) \psi, \\ \psi_{\eta}(\theta) &\equiv (2 - 5 \cos^2 \theta + 5 \cos^4 \theta)(1 - \psi) \\ &\quad + \frac{27}{2} \cos^2 \theta (1 - \cos^2 \theta) \psi, \\ \psi_{\zeta}(\theta) &\equiv \frac{1}{6} (3 + 6 \cos^2 \theta - 7 \cos^4 \theta)(1 - \psi) \\ &\quad + \frac{1}{4} (3 - 6 \cos^2 \theta + 7 \cos^4 \theta) \psi, \end{aligned} \quad (3)$$

with $\psi \equiv 1$ for $Q \parallel [001]$ and $\psi \equiv 0$ for $Q \parallel [111]$. In obtaining eq. (2), we have neglected the

terms in $K_1 \cos(4QZ)$ and $K_1 \sin(4QZ)$ for $Q \parallel [001]$ and the terms in $K_1 \cos(3QZ)$ and $K_1 \sin(3QZ)$ for $Q \parallel [111]$. The equilibrium condition of θ is obtained by minimizing $\mathcal{H}$ given by eq. (2) under $S_{\zeta}(r) \equiv S$ and $S_{\xi}(r) = S_{\eta}(r) = 0$ as follows:

$$\frac{H}{S} \cos \theta = \frac{D^2}{c} \cos^2 \theta + K_1 S^2 [\psi_{\xi}(\theta) - 2\psi_{\zeta}(\theta)]. \quad (4)$$

The equilibrium value of S also depends on H .¹²⁾ However, we do not go into detail of the H dependence of S , since the H dependence is too weak to affect the qualitative properties of the spin wave. The critical field H_c satisfies

$$\frac{H_c}{S} = \frac{D^2}{c} + K_1 S^2 [\psi_{\xi}(0) - 2\psi_{\zeta}(0)]. \quad (5)$$

Since the AEI is much weaker than the SEI in usual compounds, $H_c (\simeq D^2 S/c)$ is very weak,¹²⁾ for example, about $6g\mu_B \cdot \text{kOe}$ in MnSi.¹⁸⁻²⁰⁾ Therefore, demagnetization fields are not necessarily weak enough to be neglected compared with H_c . Nevertheless, the demagnetization field effects have been dropped from eqs. (1) and (2) in order to see essential features of the spin wave. More detailed analyses taking account of the demagnetization fields will be given in §4 and Appendix II.

2.2 Spin waves

As in the usual way, $S(r)$ is expressed in terms of the macroscopic, free spin wave operators $a(r)$ and $a(r)^+$, which satisfy $[a(r), a(r')^+] = \delta(r - r')$:

$$\begin{aligned} S_{\xi}(r) &= (S/2)^{1/2} [a(r) + a(r)^+], \\ S_{\eta}(r) &= -i(S/2)^{1/2} [a(r) - a(r)^+], \\ S_{\zeta}(r) &= S - a(r)^+ a(r). \end{aligned} \quad (6)$$

Furthermore, the Fourier transform of $a(r)$ is defined by²²⁾

$$a(r) = \sum_q a_q \exp(iq \cdot r). \quad (7)$$

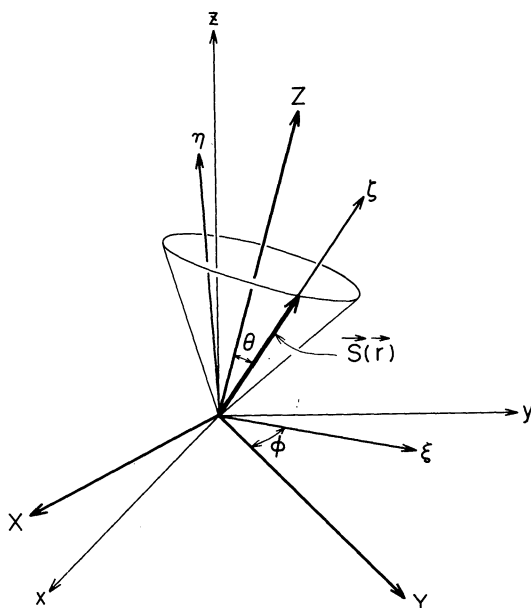


Fig. 1. The three coordinates (x, y, z) , (X, Y, Z) , and (ξ, η, ζ) . The x , y , and z axes are parallel to the cubic axes. The Z axis is parallel to the direction of Q or the bulk magnetization and the Y axis is in the xy plane. The ζ axis is parallel to the static component of a local spin density vector $S(r)$ and the ξ axis is in the XY plane. The angles θ and $\phi (= Q \cdot r)$ are the polar angles of the ζ axis in the coordinate (X, Y, Z) .

Substituting eqs. (6) and (7) into eq. (2), we obtain

$$\mathcal{H} = \sum_q [\varepsilon_q a_q^+ a_q + \frac{1}{2} \Delta_q (a_q a_{-q} + a_{-q}^+ a_q^+) + (V_q a_q^+ a_{q+Q} + V_q^* a_{q+Q}^+ a_q)], \quad (8)$$

with

$$\varepsilon_q = S \left\{ cq^2 - \frac{D^2}{2c} (2 - 3 \sin^2 \theta) - \frac{1}{2} K_1 S^2 [\psi_\xi(\theta) + \psi_\eta(\theta) - 4\psi_\zeta(\theta)] + \frac{H}{S} \cos \theta \right\}, \quad (9)$$

$$\Delta_q = -S \left\{ \frac{D^2}{2c} \sin^2 \theta + \frac{1}{2} K_1 S^2 [\psi_\xi(\theta) - \psi_\eta(\theta)] \right\}, \quad (10)$$

$$V_q = V_{q \pm Q} = SD(q_x - iq_y) \sin \theta. \quad (11)$$

Although Δ_q is independent of q in eq. (10), Δ_q is in general a function of q as will be seen later. The free spin wave with a general q couples to those with $-q$ and $q \pm nQ$, n being an integer. When a four wave approximation is employed, eq. (8) can be diagonalized to give the two low-lying spin wave energies as

$$\begin{pmatrix} \hbar\omega_q^2 \\ \hbar\omega_{q'}^2 \end{pmatrix} = \frac{1}{2} \{ [(\varepsilon_q^2 + \varepsilon_{q-Q}^2) + 2(|V_q|^2 - \Delta_q^2)] \pm [(\varepsilon_q^2 - \varepsilon_{q-Q}^2)^2 + 4|V_q|^2((\varepsilon_q + \varepsilon_{q-Q})^2 - 4\Delta_q^2)]^{1/2} \}. \quad (12)$$

Here, $q-Q$ means the one with the smaller magnitude among $q-Q$ and $q+Q$.

When q is parallel to Q , the free spin waves with q and $q-Q$ no longer mix with each other because of $V_q=0$ in our Hamiltonian. In this case, eq. (12) is reduced by use of eq. (4) to a simple form as

$$\hbar\omega_{q \parallel Q} = S \sqrt{c} |q_z| \left\{ cq_z^2 + \frac{D^2}{c} \sin^2 \theta + K_1 S^2 [\psi_\xi(\theta) - \psi_\eta(\theta)] \right\}^{1/2}. \quad (13)$$

Therefore, the spin wave energy $\hbar\omega_q$ has no gap at $q=0$ and no anomaly at $q=\pm Q$ even in the CSDW phase with the cubic magnetic anisotropy. When q is deviated from the direction of Q , $\hbar\omega_q$ has gaps at the planes $q_z = \pm Q/2$, which correspond to the new Brillouin zone boundaries in the long-period spin structure.^{13,14} Figure 2 (a) shows

schematically the spin wave dispersion curve in the HSDW phase under $H=0$. Comparing this curve with that in the case of the SEI shown in Fig. 2 (b) which will be discussed in the next section, we see a large difference between the spin wave dispersions in the two types of HSDW. When H reaches H_c , V_q vanishes (as does Δ_q), so that Q can be no longer a reciprocal lattice vector of the spin system and $\hbar\omega'_q (= \varepsilon_{q-Q})$ becomes fictitious. By use of eqs. (12) and (5), $\hbar\omega_q$ at $H=H_c-0$ is expressed by²¹⁾

$$\hbar\omega_q = Scq^2. \quad (14)$$

This dispersion is again very different from that in the case of the SEI as shown in Fig. 3.

In the IFMS above H_c , the Z axis itself becomes parallel to H and the static magnetization. $\mathcal{H}$ given by eq. (1) is changed effectively to

$$\mathcal{H} = \int d\mathbf{r} \left\{ \frac{c}{2} \sum_j \left[\left(\frac{\partial S_x(\mathbf{r})}{\partial r_j} \right)^2 + \left(\frac{\partial S_y(\mathbf{r})}{\partial r_j} \right)^2 \right] + D \left[S_x(\mathbf{r}) \left(\frac{\partial S_y(\mathbf{r})}{\partial Z} \right) - S_y(\mathbf{r}) \left(\frac{\partial S_x(\mathbf{r})}{\partial Z} \right) \right] - \frac{1}{2} K_1 [\psi_\xi(0) S_x(\mathbf{r})^2 + \psi_\eta(0) S_y(\mathbf{r})^2 + \psi_\zeta(0) S_z(\mathbf{r})^2] S_z(\mathbf{r})^2 - HS_z(\mathbf{r}) \right\}. \quad (15)$$

Since this $\mathcal{H}$ does not contain $\cos(QZ)$ or $\sin(QZ)$, the free spin wave with q no longer couples to those with $q \pm nQ$. Equations (15),

(6), and (5) yield the spin wave energy in the IFMS as

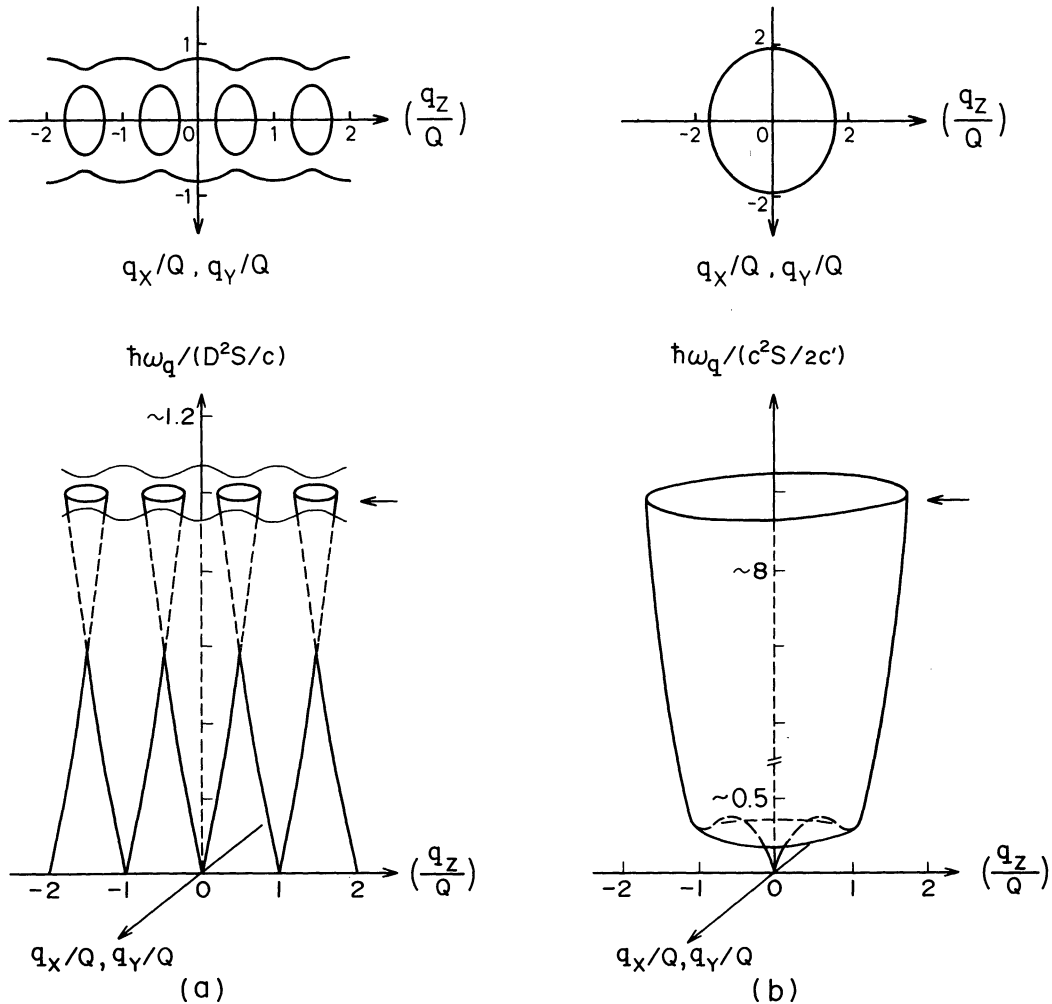


Fig. 2. (a) The spin wave energy $\hbar\omega_q$ in the helical spin density wave (HSDW) state due to the antisymmetric exchange interaction (AEI). The upper figure shows a cross section at a fixed $\hbar\omega_q$ indicated by the arrow in the lower figure. The energy gaps appear in the plane $q_z = nQ/2$ with any integer n except at $q = nQ/2$. (b) The spin wave energy $\hbar\omega_q$ in the HSDW state due to the symmetric exchange interaction (SEI). The upper figure shows a cross section at a fixed $\hbar\omega_q$ indicated by the arrow in the lower figure. At $q = |Q|$, $\hbar\omega_q$ has local minimum values which become zero in the absence of the magnetic anisotropy energy. The dispersion is anisotropic because of the presence of the long-period spin structure.

$$\hbar\omega_q = Sc|q - Q|^2 + (H - H_c), \quad (16)$$

where Q is parallel to H and Q equals $-D/c$. This dispersion curve is shown in Fig. 4 (a) for comparison with that in the case of the SEI. As seen from this figure, the curve crosses the $q=0$ axis with a nonzero slope, i.e., $(d\omega_q/dq)_{q=0} \neq 0$, except in the direction of q perpendicular to Q . This is a characteristic of the presence of the AEI. At $H = H_c + 0$, $\hbar\omega_q$ becomes completely soft but $\hbar\omega_{-Q}$ does not ex-

hibit any softening.²²⁾ This result verifies the occurrence of only one of the clockwise and counterclockwise rotations of spins in the HSDW due to the AEI. The spin wave dispersions in the IFMS and in the CSDW state do not coincide with each other at $H = H_c$. This is because a wave vector Q of a spin wave in the IFMS is used for the reciprocal lattice vector in the CSDW phase and the equilibrium condition (4) guarantees $\hbar\omega_q = 0$ at $q = 0$ in the CSDW phase. However, the intensity distribu-

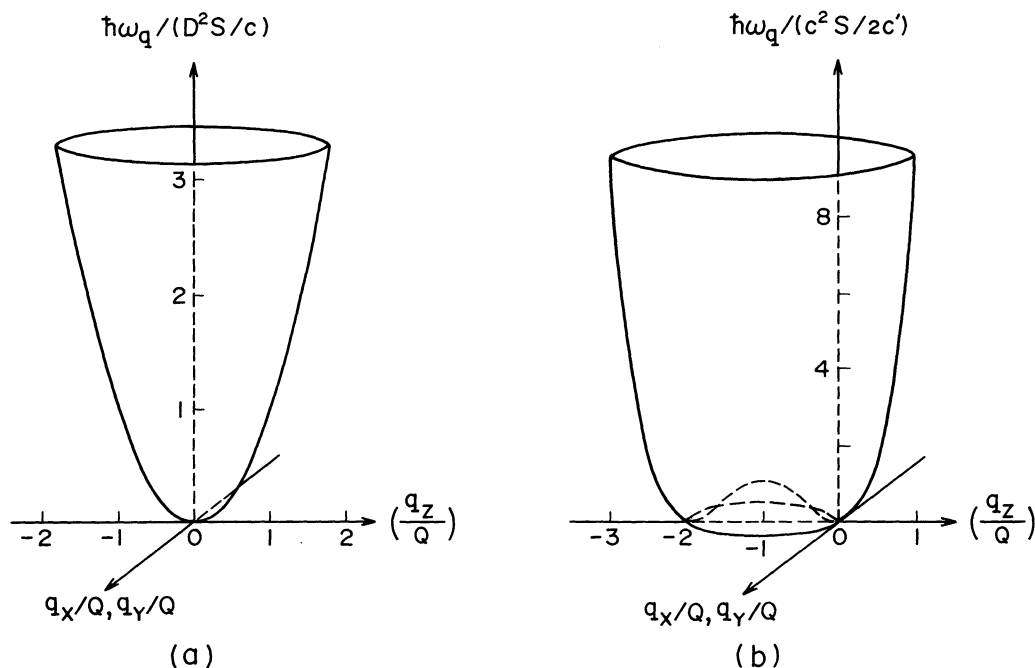


Fig. 3. (a) The spin wave energy $\hbar\omega_q$ in the conical spin density wave (CSDW) state due to the AEI at $H=H_c-0$. In spite of the presence of the magnetic anisotropy energy, $\hbar\omega_q$ becomes parabolic. (b) The spin wave energy $\hbar\omega_q$ in the CSDW state due to the SEI at $H=H_c-0$. In spite of the presence of the magnetic anisotropy energy, $\hbar\omega_q$ becomes quartic.

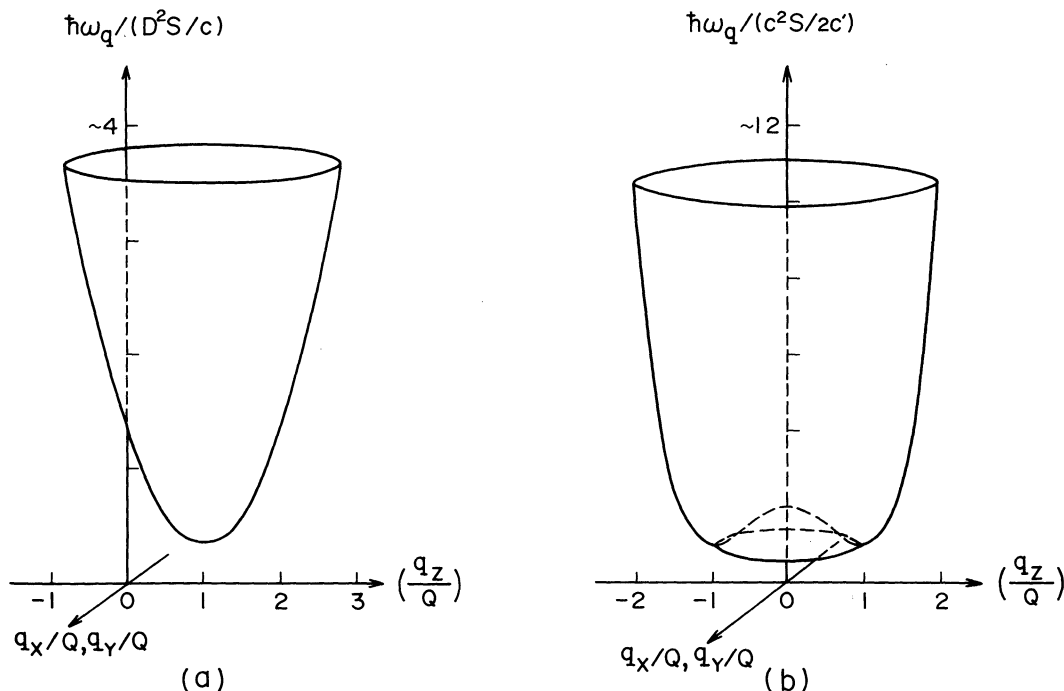


Fig. 4. (a) The spin wave energy $\hbar\omega_q$ in the induced ferromagnetic state (IFMS) at $H>H_c$ in the case of the AEI. $\hbar\omega_q$ exhibits only a single minimum, and has nonvanishing slopes at $q=0$ except in directions perpendicular to $\mathbf{Q}$. (b) The spin wave energy $\hbar\omega_q$ in the IFMS at $H>H_c$ in the case of the SEI. $\hbar\omega_q$ exhibits double minima in any direction of q .

tion of inelastic neutron scattering by spin waves does not exhibit any discontinuous change at $H=H_c$.

§3. Spin Waves in the System with the Helical Spin Density Wave (HSDW) Due to the Symmetric Exchange Interaction (SEI)

3.1 Phenomenological Hamiltonian and ground spin structure

The spin wave in this case was already in-

vestigated by Yosida and Miwa²³⁾ and others^{24,25)} for localized spin systems. In order to make clear the difference between the two types of HSDW, we consider the spin system in which the spin density varies slowly with r and the HSDW is stabilized without the AEI. The phenomenological Hamiltonian of a cubic crystal with a unit volume is assumed to be

$$\mathcal{H} = \int d\mathbf{r} \left\{ \frac{c}{2} \sum_{ii'} \left(\frac{\partial S_i(\mathbf{r})}{\partial r_{i'}} \right)^2 + \frac{c'}{4} \sum_{ii'} \left(\frac{\partial^2 S_i(\mathbf{r})}{\partial r_{i'}^2} \right) \left(\frac{\partial^2 S_i(\mathbf{r})}{\partial r_{i'}^2} \right) - \frac{1}{2} K_1 \sum_i S_i(\mathbf{r})^4 - \mathbf{H} \cdot \mathbf{S}(\mathbf{r}) \right\}. \quad (17)$$

In this equation, the SEI, which is represented by the first and second terms, has been approximated again to be isotropic with respect to the directions of derivatives. Here, we assume $c < 0$, $c' > 0$ and a sufficiently small $|K_1|$, for which a HSDW can occur without the aid of the AEI. After transforming the coordinate as in the preceding section, eq. (17) is changed effectively to be

$$\begin{aligned} \mathcal{H} = \int d\mathbf{r} \left\{ \frac{c}{2} \sum_j \left[\left(\frac{\partial S_\xi(\mathbf{r})}{\partial r_j} \right)^2 + \left(\frac{\partial S_\eta(\mathbf{r})}{\partial r_j} \right)^2 \right] + \frac{c'}{4} \left[\left(\sum_j \frac{\partial^2 S_\xi(\mathbf{r})}{\partial r_j^2} \right)^2 + \left(\sum_j \frac{\partial^2 S_\eta(\mathbf{r})}{\partial r_j^2} \right)^2 \right] \right. \\ + \left(\frac{c}{2} + \frac{c'}{4} Q^2 \right) Q^2 [S_\xi(\mathbf{r})^2 + \cos^2 \theta S_\eta(\mathbf{r})^2 + \sin^2 \theta S_\zeta(\mathbf{r})^2] \\ + c' Q^2 \sum_j \left(\frac{1}{2} + \delta_{jz} \right) \left[\left(\frac{\partial S_\xi(\mathbf{r})}{\partial r_j} \right)^2 + \cos^2 \theta \left(\frac{\partial S_\eta(\mathbf{r})}{\partial r_j} \right)^2 \right] \\ + c' Q \cos \theta \sum_j \left[\left(\frac{\partial S_\xi(\mathbf{r})}{\partial Z} \right) \left(\frac{\partial^2 S_\eta(\mathbf{r})}{\partial r_j^2} \right) - \left(\frac{\partial S_\eta(\mathbf{r})}{\partial Z} \right) \left(\frac{\partial^2 S_\xi(\mathbf{r})}{\partial r_j^2} \right) \right] \\ \left. - \frac{1}{2} K_1 [\psi_\xi(\theta) S_\xi(\mathbf{r})^2 + \psi_\eta(\theta) S_\eta(\mathbf{r})^2 + \psi_\zeta(\theta) S_\zeta(\mathbf{r})^2] S_\zeta(\mathbf{r})^2 - H S_\zeta(\mathbf{r}) \cos \theta \right\}, \quad (18) \end{aligned}$$

where $\mathbf{H}$ has been applied along $\mathbf{Q}$ and δ_{jz} means 1 for $j=Z$ and 0 for $j=X$ and Y . The energy E of the static spin structure with $S_\xi(\mathbf{r}) \equiv S$ and $S_\zeta(\mathbf{r}) = S_\eta(\mathbf{r}) = 0$ has a simple form as

$$\begin{aligned} E = \left(\frac{1}{2} c Q^2 + \frac{1}{4} c' Q^4 \right) S^2 \sin^2 \theta \\ - \frac{1}{2} K_1 \psi_\zeta(\theta) S^4 - H S \cos \theta. \quad (19) \end{aligned}$$

This equation yields the equilibrium conditions of the CSDW which are given by

$$Q^2 = -c/c', \quad (20)$$

$$\frac{H}{S} \cos \theta = \frac{c^2}{2c'} \cos^2 \theta + K_1 S^2 [\psi_\zeta(\theta) - 2\psi_\zeta(\theta)], \quad (21)$$

together with $2c^2/c' > K_1 S^2 > 0$ for $\mathbf{Q} \parallel [001]$

and $3c^2/c' > -K_1 S^2 > 0$ for $\mathbf{Q} \parallel [111]$. The positive $Q (= \sqrt{|c|/c'})$ and negative $Q (= -\sqrt{|c|/c'})$ again correspond, respectively, to the counterclockwise and clockwise rotations of $\mathbf{S}(\mathbf{r})$. However, eq. (20) does not determine the sign of Q , allowing both rotations of spins. It is noted that eq. (21) agrees with eq. (4) if $(c^2/2c')$ is replaced by (D^2/c) in eq. (21). This implies the qualitatively same magnetization process in the two types of HSDW. From eq. (21), H_c in this case is expressed by

$$\frac{H_c}{S} = \frac{c^2}{2c'} + K_1 S^2 [\psi_\zeta(0) - 2\psi_\zeta(0)]. \quad (22)$$

When $H_c (\simeq c^2 S^2 / 2c')$ is sufficiently small, effects of the demagnetization field cannot be neglected, being discussed in the next section

and Appendix II.

linear approximation for the spin motion is made to result in

3.2 Spin waves

After substitution of eq. (6) into eq. (18), a

$$\mathcal{H} = \sum_q \left[(\varepsilon_q + 2Sc' Q q^2 q_z \cos \theta) a_q^+ a_q + \frac{1}{2} \Delta_q (a_q a_{-q} + a_{-q}^+ a_q^+) \right], \quad (23)$$

with

$$\varepsilon_q = S \left\{ \left(cq^2 + \frac{c'}{2} q^4 \right) - \frac{c^2}{4c'} (2 - 3 \sin^2 \theta) - \frac{c}{2} (q^2 + 2q_z^2) (1 + \cos^2 \theta) - \frac{1}{2} K_1 S^2 [\psi_\xi(\theta) + \psi_\eta(\theta) - 4\psi_\zeta(\theta)] + \frac{H}{S} \cos \theta \right\}, \quad (24)$$

$$\Delta_q = -S \left\{ \frac{c^2}{4c'} \sin^2 \theta + \frac{c}{2} (q^2 + 2q_z^2) \sin^2 \theta + \frac{1}{2} K_1 S^2 [\psi_\xi(\theta) - \psi_\eta(\theta)] \right\}. \quad (25)$$

In our assumed Hamiltonian (18), the free spin waves with different magnitudes of q no longer mix with each other so that the Brillouin zone of the long-period spin structure is remained to be unchanged. Equation (23) is easily diagonalized to give

$$\hbar\omega_q = (\varepsilon_q^2 - \Delta_q^2)^{1/2} + 2Sc' Q q^2 q_z \cos \theta. \quad (26)$$

When q is parallel to Q , $\hbar\omega_q$ is expressed by use of eq. (21) by

$$\hbar\omega_{q \parallel Q} = S \left\{ |q_z| \left[2|c| + \frac{c'}{2} q_z^2 \right]^{1/2} \left[\left(cq_z^2 + \frac{c'}{2} q_z^4 \right) + \frac{c^2}{2c'} \sin^2 \theta - 3c q_z^2 \cos^2 \theta + K_1 S^2 (\psi_\xi(\theta) - \psi_\eta(\theta)) \right]^{1/2} + 2c' Q q_z^3 \cos \theta \right\}. \quad (27)$$

The spin wave energy has again no energy gap at $q=0$ in the presence of the cubic magnetic anisotropy. However, $\hbar\omega_q$ under $H=0$ has another local minimum value proportional to $|K_1|^{1/2}$ at $q = \pm Q$. Figure 2 (b) shows $\hbar\omega_q$ in the HSDW state due to the SEI, being compared with Fig. 2 (a). The appearance of the long-period spin structure makes the spin wave dispersion anisotropic. At a sufficiently small q , for example, $\hbar\omega_{q \perp Q}$ is quadratic in q_x or q_y while $\hbar\omega_{q \parallel Q}$ is linear in $|q_z|$ as seen from eq. (27). When H is increased up to H_c , the dispersion curve is shifted toward the direction of $-Q$, preserving the absence of the energy gap at $q=0$. At $H=H_c-0$, eq. (26) becomes

$$\hbar\omega_q = \frac{1}{2} Sc' (q^2 + 2Qq_z)^2. \quad (28)$$

This $\hbar\omega_q$ is shown in Fig. 3 (b) for comparison with that in the case of the AEI. It should be noted that $\hbar\omega_q$ is expressed by a quartic equation within a wave vector region interested here in spite of the presence of the magnetic anisotropy. Our Hamiltonian gives $\hbar\omega_q=0$ on the sphere given by $q_x^2 + q_y^2 + (q_z + Q)^2 = Q^2$ at $H=H_c$. If an anisotropic spatial dependence of the SEI is taken into account in eq. (17), $\hbar\omega_q$ becomes anisotropic around $q=-Q$. However, its double minima structure is unchanged.

In the IFMS with $Z \parallel H$ above H_c , eq. (17) is effectively written as

$$\mathcal{H} = \int d\mathbf{r} \left\{ \frac{c}{2} \sum_j \left[\left(\frac{\partial S_x(\mathbf{r})}{\partial r_j} \right)^2 + \left(\frac{\partial S_y(\mathbf{r})}{\partial r_j} \right)^2 \right] + \frac{c'}{4} \left[\left(\sum_j \frac{\partial^2 S_x(\mathbf{r})}{\partial r_j^2} \right)^2 + \left(\sum_j \frac{\partial^2 S_y(\mathbf{r})}{\partial r_j^2} \right)^2 \right] \right\}$$

$$-\frac{1}{2} K_1 [\psi_\xi(0) S_X(r)^2 + \psi_\eta(0) S_Y(r)^2 + \psi_\zeta(0) S_Z(r)^2] S_Z(r)^2 - H S_Z(r) \Big\}. \quad (29)$$

Equations (29), (6), and (22) result in

$$\hbar\omega_q = \frac{1}{2} S c' (q^2 - Q^2)^2 + (H - H_c). \quad (30)$$

It is noted that $\hbar\omega_q$ depends on $Q^2 (= -c/c')$ but not on a direction of $\mathbf{Q}$, making a contrast to $\hbar\omega_q$ in the case of the AEI. As shown in Fig. 4 (b), this dispersion curve has double minima in any direction of $\mathbf{q}$. When, therefore, H is decreased to H_c , $\hbar\omega_q$ can be completely softened at $\mathbf{q} = \pm \mathbf{Q}$, which correspond to clockwise and counterclockwise rotations of spins below H_c . Since $\hbar\omega_q$ given by eq. (30) is an even function of $\mathbf{q}$, it has no slope at $\mathbf{q} = 0$, i.e., $(d\omega_q/d\mathbf{q})_{\mathbf{q}=0} = 0$. These make a sharp contrast with the properties in the case of the AEI. Also in the SEI case, the spin wave dispersions in the IFMS and in the CSDW state do not coincide with each other at $H = H_c$ by reasons similar to those in the AEI case.

§4. Electron Spin Resonance (ESR) Frequencies of the Systems with the Helical Spin Density Waves (HSDW) Due to the Antisymmetric Exchange Interaction (AEI) and the Symmetric Exchange Interaction (SEI)

By the same reason proved by Cooper *et*

al.,¹⁶⁾ an oscillating magnetic field, $\mathbf{H}_{rf}$, applied to the CSDW states can excite the spin wave with $\mathbf{q} = 0$ for $\mathbf{H}_{rf} \parallel \mathbf{Q}$ and the spin waves with $\mathbf{q} = \pm \mathbf{Q}$ for $\mathbf{H} \perp \mathbf{Q}$, irrespective to the origins of the HSDW's. (See also Appendix III.) In this states, $\mathbf{H}_{rf}$ is assumed to be perpendicular to $\mathbf{Q}$, since $\hbar\omega_{\mathbf{q}=0}$ is 0 as was shown in the preceding sections. For real materials, our above calculations should be improved by taking account of the demagnetization field. As a representative case, we here consider spherical shapes of crystals, in which the demagnetization field from the static magnetization vanishes. (The spin wave energy in a more general shape of crystal is discussed in Appendix II.) Even in a spherical sample, a spin wave with a general $\mathbf{q}$ produces a time-dependent, local demagnetization field. When $\mathbf{q}$ is parallel to $\mathbf{Q}$, the excess energy originating from the demagnetization field from the spin waves, $\mathcal{H}_d^{SW}$, is given by

$$\begin{aligned} \mathcal{H}_d^{SW} &= 2\pi(g\mu_B)^2 \sin^2\theta \int S_\eta(r)^2 dr \\ &= 2S\pi(g\mu_B)^2 \sin^2\theta \sum_{\mathbf{q} \parallel \mathbf{Q}} \left[a_q^+ a_q - \frac{1}{2} (a_q a_{-q} + a_{-q}^+ a_q^+) \right]. \end{aligned} \quad (31)$$

The AEI case Diagonalization of $\mathcal{H}$ given by eq. (8) plus $\mathcal{H}_d^{SW}$ yields the ESR energy given by

$$\hbar\omega_{\pm Q} = S \left\{ \left(\frac{D^2}{c} \right) \left[\left(\frac{D^2}{c} \right) + \left(\frac{D^2}{c} + 4\pi g^2 \mu_B^2 \right) \sin^2\theta + K_1 S^2 (\psi_\xi(\theta) - \psi_\eta(\theta)) \right] \right\}^{1/2}, \quad (32)$$

where eq. (5) has been used. When $|K_1|$ is small enough to be neglected, this equation is reduced to²⁶⁾

$$\begin{aligned} \hbar\omega_{\pm Q} &= H_c \left\{ 1 + \left(1 + \frac{4\pi g^2 \mu_B^2 S}{H_c} \right) \left[1 - \left(\frac{H}{H_c} \right)^2 \right] \right\}^{1/2}. \end{aligned} \quad (33)$$

Our assumed $\mathcal{H}$ given by eq. (8) does not split $\hbar\omega_{+Q}$ and $\hbar\omega_{-Q}$. Even in the case of the AEI, however, $\mathcal{H}$ contains in general the energy represented by the second term in eq. (17), which can lift their degeneracy. After a similar calculation to that in the SEI case which has been argued in detail in the preceding section, the splitting is proved to be given by

$$\hbar\omega_{+q} - \hbar\omega_{-q} = 4Sc'(D/c)^4 \cos \theta. \quad (34)$$

This is estimated to be of the order of $H_c(|\mathbf{Q}|/|\mathbf{G}|)^2 \cos \theta$, $\mathbf{G}$ being a reciprocal lattice vector. Since $(|\mathbf{Q}|/|\mathbf{G}|)^2 \ll 1$ is expected in the AEI case, the splitting is much small compared with $\hbar\omega_{\pm q}$ itself. In the IFMS above H_c , the ferromagnetic resonance energy $\hbar\omega_{q=0}$ is

obtained from eq. (16) as

$$\hbar\omega_{q=0} = H - K_1 S^3 [\psi_\zeta(0) - 2\psi_\zeta(0)]. \quad (35)$$

The magnetic field dependence of the resonance energy in this case is shown in Fig. 5 (a).

The SEI case $\mathcal{H}$ given by eq. (23) plus $\mathcal{H}_d^{\text{SW}}$ is diagonalized to give the resonance energy in the CSDW phase as

$$\begin{pmatrix} \hbar\omega_{+q} \\ \hbar\omega_{-q} \end{pmatrix} = S \left(\frac{c^2}{2c'} \right) \left\{ 5 \left[1 + \left(-1 + \frac{8\pi c' g^2 \mu_B^2}{5c^2} \right) \sin^2 \theta + \frac{2c' K_1 S^2}{5c^2} (\psi_\zeta(\theta) - \psi_\eta(\theta)) \right]^{1/2} \pm 4 \cos \theta \right\}. \quad (36)$$

When $|K_1|$ can be neglected, $\hbar\omega_{+q}$ and $\hbar\omega_{-q}$ have the magnetic field dependencies as follows:

$$\begin{pmatrix} \hbar\omega_{+q} \\ \hbar\omega_{-q} \end{pmatrix} = H_c \left\{ 5 \left[1 - \left(1 - \frac{4\pi g^2 \mu_B^2 S}{5H_c} \right) \left(1 - \frac{H^2}{H_c^2} \right) \right]^{1/2} \pm 4 \left(\frac{H}{H_c} \right) \right\}. \quad (37)$$

In contrast to the case of the AEI, the splitting of $\hbar\omega_{+q}$ and $\hbar\omega_{-q}$ becomes much large at $H=H_c$ compared with $\hbar\omega_{\pm q}$ at $H=0$, even for a sufficiently small $|\mathbf{Q}|$. In the IFMS above H_c , the ferromagnetic resonance energy is expressed by eq. (35) also in this case. The magnetic field dependence of the resonance

energy in the SEI case is shown in Fig. 5 (b), which should be compared with Fig. 5 (a).

In both cases, however, the cubic magnetic anisotropy energy mixes spin waves with $\mathbf{q}$ and $\mathbf{q} \pm 4\mathbf{Q}$ for $\mathbf{Q} \parallel [001]$ and those with $\mathbf{q}$ and $\mathbf{q} \pm 3\mathbf{Q}$ for $\mathbf{Q} \parallel [111]$ (See §2), and furthermore the energy of the demagnetization field from

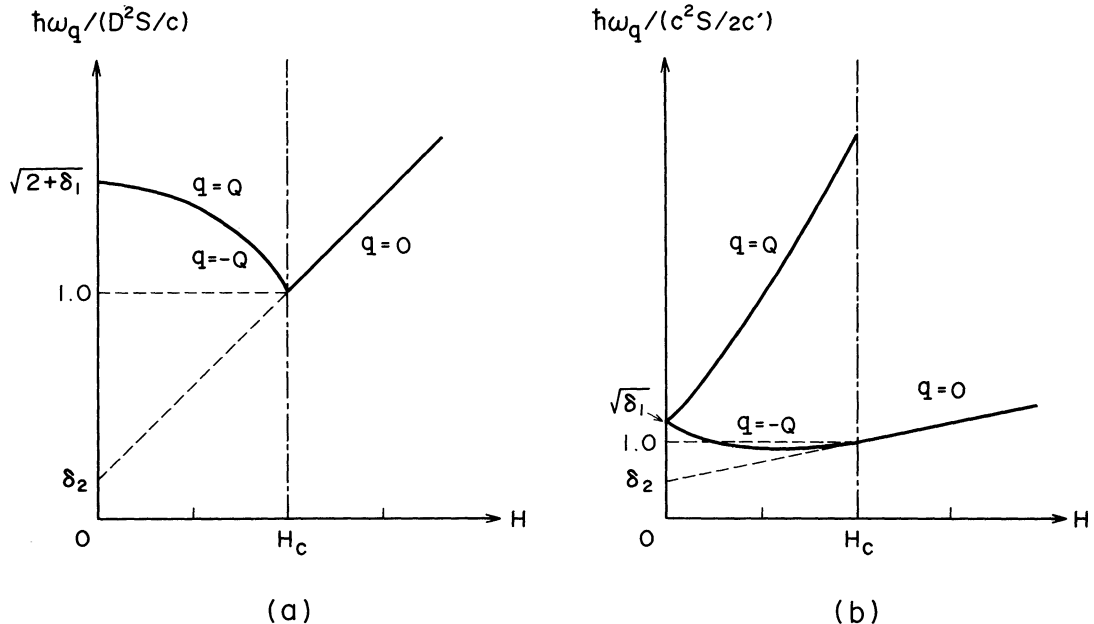


Fig. 5. (a) Magnetic field dependence of the electron spin resonance (ESR) energy in the case of the HSDW due to the AEI. The splitting of $\hbar\omega_{+q}$ and $\hbar\omega_{-q}$ at $H \neq 0$ is small. The positive constants δ_1 and δ_2 are known from eqs. (32) and (35). (b) Magnetic field dependence of the ESR energy in the case of the SEI. The splitting of $\hbar\omega_{+q}$ and $\hbar\omega_{-q}$ at $H \neq 0$ is large. δ_1 and δ_2 are known from eqs. (36) and (35).

the static magnetization mixes spin waves with q and $q \pm 2Q$. (See Appendix II). Therefore, the spin wave energy can have, generally speaking, many gaps even at $q = \pm Q$, which allow new resonance energies in the CSDW phases. A somewhat important situation comes out in the AEI case: The nearly degenerate $\hbar\omega_{+Q}$ and $\hbar\omega_{-Q}$ are split by the coupling of the spin waves Q and $-Q$ through the demagnetization field. This splitting does not vanish even at $H=0$, differing from that given by eq. (34).

As seen from Fig. 5, the magnetic field dependences of the resonance energies in the two types of HSDW are different enough to be able to distinguish the origins of HSDW's in real crystals. Date *et al.*¹⁷⁾ observed ESR in the CSDW phase and the IFMS of MnSi. Their results were analyzed on the standpoint that the HSDW is due to the SEI. In their analysis, however, a uniaxial magnetic anisotropy whose origin is not necessarily obvious was required. Contrary to their analysis, the observed magnetic field dependence is rather in agreement with that calculated for the HSDW due to the AEI, although a more detailed analysis which takes into account also effects of the demagnetization field is necessary. On the other hand, Watanabe *et al.*⁶⁾ observed ESR in $\text{Fe}_x\text{Co}_{1-x}\text{Si}$ ($0.3 \leq x \leq 0.85$). They found that the relation $\hbar\omega_{\pm Q} = \sqrt{2}H_c$, which is derived from eq. (33) with $H=0$ by neglecting the effect of the demagnetization field, does well hold in both $\text{Fe}_{0.8}\text{Co}_{0.2}\text{Si}$ and MnSi. This gives one of the evidences of the HSDW due to the AEI in these compounds.

§5. Concluding Remarks

In the preceding sections, we have seen the strongly different spin wave spectra between the HSDW's due to the AEI and the SEI. The difference becomes more conspicuous when the HSDW's are distorted to the CSDW's and squeezed to the IFMS's by external magnetic fields. Among these differences, the followings are remarked: (1) In the CSDW due to the AEI, new Brillouin zone boundaries arise at the planes $q_z = (2n+1)Q/2$ (except at the points $q = (2n+1)Q/2$). This makes a contrast to the case of the SEI. (2) In the HSDW state due to the SEI, the spin wave energy exhibits local minima at $q = \pm Q$, which can not be

found in the HSDW state due to the AEI. (3) The spin wave energy in the IFMS has a single minimum at $q=Q$ in the case of the AEI, while double minima at $q = \pm Q$ in the case of the SEI. This difference reflects the fact that only one of the clockwise and counter-clockwise rotations is allowed in the HSDW due to the AEI, but both rotations are allowed in the HSDW due to the SEI.

The large difference between the spin wave dispersions of the two different kinds of HSDW can be expected always in any other types of crystal. Unfortunately, intensities of neutron inelastic scattering by spin waves can be hardly measured with high accuracies enough to clarify the spin wave dispersions in a region of $|q|$ comparable with $|Q|$ much smaller than $|G|$. Nevertheless, ESR experiments can reveal the energies of the spin waves with $q=0$, and $\pm Q$ in the HSDW or CSDW phases, being an excellent method to determine the origin of a HSDW. In crystals with low symmetries, Q of the HSDW due to the AEI is allowed to have only some special directions in crystals.¹²⁾ Such a HSDW can exhibit a characteristic magnetization process under external magnetic fields perpendicular to Q , because a fan structure is forbidden in the HSDW due to the AEI. In fact, the HSDW with $|Q|=0.013 \text{ \AA}^{-1}$ in $\text{Cr}_{1/3}\text{NbS}_2$ was determined to be due to the AEI from the magnetization process.⁹⁾ Generally speaking, however, the static properties including the magnetization process often exhibit similar behaviours in both types of HSDW. It should be emphasized here that the difference in dynamical properties of the two types of HSDW is essential in any crystal. This further suggests that the AEI and the SEI which cooperate in crystals with low symmetries can be separated by measuring spin wave spectra.

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Appendix I

We here give the relations between the three coordinates which are shown in Fig. 1. When

$\mathbf{Q}$ is parallel to the z axis, the coordinate (X , Y , Z) coincides with the coordinate (x , y , z).

When $\mathbf{Q}$ is parallel to $[111]$ in the coordinate (x , y , z), (X , Y , Z) is defined by

$$\begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 1/\sqrt{2}, & 1/\sqrt{6}, & 1/\sqrt{3} \\ -1/\sqrt{2}, & 1/\sqrt{6}, & 1/\sqrt{3} \\ 0, & -2/\sqrt{6}, & 1/\sqrt{3} \end{pmatrix} \begin{pmatrix} X \\ Y \\ Z \end{pmatrix}. \quad (\text{A} \cdot 1)$$

The coordinate (ξ , η , ζ), on the other hand, is related to the coordinate (X , Y , Z) by

$$\begin{pmatrix} X \\ Y \\ Z \end{pmatrix} = \begin{pmatrix} -\sin \varphi, & -\cos \theta \cos \varphi, & \sin \theta \cos \varphi \\ \cos \varphi, & -\cos \theta \sin \varphi, & \sin \theta \sin \varphi \\ 0, & \sin \theta, & \cos \theta \end{pmatrix} \begin{pmatrix} \xi \\ \eta \\ \zeta \end{pmatrix}, \quad (\text{A} \cdot 2)$$

where θ and $\varphi(=\mathbf{Q} \cdot \mathbf{r})$ are the polar angles of the ζ direction in the coordinate (X , Y , Z).

Appendix II

The spin wave energies in the CSDW phases are discussed for a nonspherical shape of a crystal with cubic symmetry. We assume an elliptic shape whose principal axes coincide with the X , Y , and Z axes and whose demagnetization field coefficients are N_X , N_Y , and N_Z . For simplicity, differences between N_X , N_Y , and N_Z are assumed to be sufficiently small compared with 4π so that distortions of the CSDW's by the demagnetization field can be neglected. The excess energy $\mathcal{H}_d^{\text{SM}}$ which is attributed to the presence of the demagnetization field of the static magnetization is expressed by

$$\begin{aligned} \mathcal{H}_d^{\text{SM}} = & -\frac{1}{4} (g\mu_B)^2 \int d\mathbf{r} \left\{ (N_X - N_Y) S_\xi(\mathbf{r})^2 + \left[2 \left(\frac{4}{3} \pi - N_Z \right) \sin^2 \theta - (N_X - N_Y) \cos^2 \theta \right] S_\eta(\mathbf{r})^2 \right. \\ & \left. + \left[2 \left(\frac{4}{3} \pi - N_Z \right) \cos^2 \theta - (N_X - N_Y) \sin^2 \theta \right] S_\zeta(\mathbf{r})^2 \right\}. \quad (\text{A} \cdot 3) \end{aligned}$$

In obtaining this equation, we have used an approximation $\cos^2(\mathbf{Q} \cdot \mathbf{y}) = \sin^2(\mathbf{Q} \cdot \mathbf{r}) = 1/2$ for any $\mathbf{r}$, which neglects the mixing between the spin waves with $\mathbf{q}$ and $\mathbf{q} \pm 2\mathbf{Q}$. The energy $\mathcal{H}_d^{\text{SW}}$ which is attributed to the presence of the demagnetization field produced by a spin wave depends on the direction of $\mathbf{q}$. For convenience, $\mathbf{q}$ is here chosen to be parallel to the Z axis, so that $\mathcal{H}_d^{\text{SW}}$ is given by eq. (31). By diagonalizing $(\mathcal{H} + \mathcal{H}_d^{\text{SM}} + \mathcal{H}_d^{\text{SW}})$, the spin wave energies are found to be again expressed by eqs. (12) and (26) but with new ε_q and Δ_q modified as follows: ε_q 's are obtained by adding

$$\begin{aligned} & \frac{1}{4} (g\mu_B)^2 S [(2N_Z - N_X - N_Y)(1 - 3 \cos^2 \theta) \\ & + 8\pi \sin^2 \theta], \quad (\text{A} \cdot 4) \end{aligned}$$

to the right hand sides of eqs. (9) and (24) and Δ_q 's are obtained by adding

$$-\frac{1}{4} (g\mu_B)^2 S [(2N_Z - N_X - N_Y) \sin^2 \theta + 8\pi \sin^2 \theta], \quad (\text{A} \cdot 5)$$

to the right hand sides of eqs. (10) and (25). For the equilibrium conditions for the CSDW phases, on the other hand, the term,

$$\frac{1}{2} (g\mu_B)^2 S (2N_Z - N_X - N_Y) \cos^2 \theta, \quad (\text{A} \cdot 6)$$

should be added to the right hand sides of eqs. (4) and (21).

Appendix III

The magnetic field dependences of the ESR absorption intensities in the CSDW phases are briefly discussed. $\mathbf{H}_{\text{rf}}$ perpendicular to $\mathbf{Q}$ causes the energy $\mathcal{H}_{\text{rf}}$ given by

$$\mathcal{H}_{\text{rf}} = \sqrt{S/8} H_{\text{rf}} [\text{li}(U_Q \alpha_Q^+ + U_{-Q} \alpha_{-Q}^+) + \text{h.c.}], \quad (\text{A} \cdot 7)$$

$$\begin{pmatrix} -U_{+Q} \\ +U_{-Q} \end{pmatrix} = u_Q(1 \mp \cos \theta) + v_Q(1 \pm \cos \theta), \quad (\text{A} \cdot 8)$$

α_Q^+ , has been defined to be expressed by $\alpha_Q^+ = u_Q \alpha_Q^+ - v_Q \alpha_{-Q}$. The resonance absorption intensity A_Q is proportional to $|U_Q|^2$.¹⁶⁾ After some manipulations, we obtain

where the creation operator of the spin wave,

$$\begin{pmatrix} A_{+Q} \\ A_{-Q} \end{pmatrix} \propto \{[(\epsilon_Q - \Delta_Q) + (\epsilon_Q + \Delta_Q) \cos^2 \theta] / (\epsilon_Q^2 - \Delta_Q^2)^{1/2} \mp 2 \cos \theta\}, \quad (\text{A} \cdot 9)$$

for both types of HSDW. Equations (A·9), (9), (10), and (4) give $A_{\pm Q}$ in the case of the HSDW due to the AEI as

$$\begin{pmatrix} A_{+Q} \\ A_{-Q} \end{pmatrix} \propto \frac{2(D^2/c) + 4\pi g^2 \mu_B^2 \sin^2 \theta + K_1 S^2 [\psi_\xi(\theta) - \psi_\eta(\theta)]}{\sqrt{D^2/c} \sqrt{(D^2/c) + [(D^2/c) + 4\pi g^2 \mu_B^2] \sin^2 \theta + K_1 S^2 [\psi_\xi(\theta) - \psi_\eta(\theta)]}} \mp 2 \cos \theta. \quad (\text{A} \cdot 10)$$

When H or $\cos \theta$ is increased, A_{+Q} decreases but A_{-Q} and $(A_{+Q} + A_{-Q})$ increase for a sufficiently small $|K_1|$. In the case of the HSDW due to the SEI, eqs. (A·9), (24), (25), and (21) give

$$\begin{pmatrix} A_{+Q} \\ A_{-Q} \end{pmatrix} \propto \frac{5(c^2/c') \cos^2 \theta + 4\pi g^2 \mu_B^2 \sin^2 \theta + K_1 S^2 [\psi_\xi(\theta) - \psi_\eta(\theta)]}{\sqrt{5c^2/2c'} \sqrt{(5c^2/2c') \cos^2 \theta + 4\pi g^2 \mu_B^2 \sin^2 \theta + K_1 S^2 [\psi_\xi(\theta) - \psi_\eta(\theta)]}} \mp 2 \cos \theta. \quad (\text{A} \cdot 11)$$

Whether A_{+Q} and/or A_{-Q} increase or decrease with increasing H depends on relative magnitudes of c^2/c' , $g^2 \mu_B^2$, and $K_1 S^2$ in this case. It is noted that A_Q equals A_{-Q} at $H=0$ in both types of HSDW.

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